

PMMR: Query-Adaptive Geo-Personalized Multimodal Ranking for Creative Search at Scale

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Abstract—We present PMMR, a production-grade, multi-stage ranking framework for creative search that combines multimodal fusion, segment-aware personalization, and geo-contextual optimization under strict latency budgets. PMMR re-ranks BM25 and ANN candidates with a query-adaptive fusion of CLIP-like and knowledge-graph signals together with behavior-derived features. The ranker is trained with a pairwise RankNet objective enhanced by soft negative mining and is evaluated offline with inverse propensity scoring and slice-aware analyses. Deployed in Adobe Express, PMMR consistently outperforms an industrial HR+RN baseline: offline gains in MRR and NDCG translate in online AB tests to higher CTR (+3.46%), improved export conversion rate (+1.15%), and a large reduction in null rate (~67%), while meeting end-to-end tail latency P99 under 400 milliseconds. Gains generalize across locales, head–torso–tail query buckets, and user segments. Beyond methodology, this work offers a transferable blueprint for large-scale search and recommendation systems that unite multimodal fusion, geo-personalization, and rigorous offline–online evaluation.

Index Terms—Multimodal, search ranking, personalization, geo-aware ranking, recommendation

I. INTRODUCTION

Creative search in platforms like Adobe Express differs from classic web IR: queries are terse, subjective (e.g., “elegant birthday card”), and multilingual, and success is defined by downstream usage (remix/export) and business KPIs rather than clicks alone. These realities demand a ranker that unifies multimodal understanding, adaptive personalization, and geo-awareness under strict latency budgets.

We present **PMMR**, a production-validated, multi-stage *Personalized Multi-Modal Ranking* framework deployed in Adobe Express. PMMR augments a hybrid BM25+ANN candidate pipeline with: (i) *dual-embedding fusion* of CLIP-like image–text and Creative Knowledge Graph (CKG) signals; (ii) *segment-aware personalization* to reflect distinct intents across individual, enterprise, and education cohorts; and (iii) *geo-personalized smoothing* using Laplace-backed regional engagement (CTR/remix/export) with category-wise normalization and conservative caps. A streamlined pairwise RankNet with soft negative mining learns component weights from de-biased click signals and supports robust re-ranking at production scale. The result is a modular, low-latency system that adapts to regional semantics (e.g., *football* in US vs. France) and cohort preferences while preserving diversity and fairness across locales.

Contributions. This paper make four contributions:

- 1) **Query-Adaptive Dual-Modal Fusion:** Joint use of CLIP-like and CKG similarities combined with BM25 via a lightweight, query-adaptive gating strategy.
- 2) **Segment Personalization:** A compact personalization head using user-segment embeddings and content/context features, improving relevance for individual/enterprise/educational cohorts.
- 3) **Geo-Personalized Optimization:** Region-aware engagement priors with Laplace backoff, category normalization, and caps to avoid popularity leakage in low-traffic locales.
- 4) **Production-Grade Learning-to-Rank:** Pairwise RankNet with soft negative mining over actual BM25/ANN neighborhoods; Offline evaluation and human side-by-side labels align with online CTR and export lifts.

Figure 1 sketches the end-to-end system. We summarize feature families in Table I; define locale frequency buckets and *Overlap@20* in Table II; report offline cumulative and ablation results in Tables IV and V; include human side-by-side outcomes in Table X; and show latency budgets in Table XIV. Our experiments connect themes of multimodal fusion, geo-personalization, and fairness, and demonstrate consistent offline/online gains under tight latency constraints.

II. RELATED WORK

Multimodal retrieval and ranking. Industrial IR systems increasingly combine sparse and dense signals; CLIP-style vision–language models and sequence architectures enable cross-modal alignment for retrieval and recommendation [12], [14], [20], [15]. Most prior work optimizes representation learning and first-pass retrieval quality, but stops short of integrating adaptive re-ranking with production constraints (latency, traffic spikes), or of coupling multimodal evidence with user/locale context at inference time. In contrast, PMMR learns to *re-score* hybrid candidates by fusing CLIP-like and creative-knowledge-graph (CKG) similarities with behavior-derived and locale-aware features under a single learned objective.

Personalized search. Personalization can improve relevance when user intent varies across cohorts [16], [17]. Prior approaches often rely on static user profiles or heuristic boosts, which underperform in creator workflows where the user intent shifts with task, device, and language. PMMR departs from static heuristics by learning *segment-aware* signals (e.g.,

individual, enterprise, educational) and multi-window behavioral statistics that regularize cold-start items while preserving diversity.

Geo-aware ranking. Location signals have been exploited for entity- or proximity-centric ranking [18], [19]. However, creative marketplaces require finer-grained regional priors (e.g., color palettes, format norms) and robust treatment of low-traffic locales. PMMR implements *geo-personalized smoothing* with Laplace backoff and caps that prevent popularity leakage from high-traffic regions, and we audit fairness slices across locales and frequency buckets.

Comparison to prior Adobe systems. Our work extends Adobe’s Smart Multi-Modal Search (hybrid BM25 + ANN fusion in Elasticsearch) [7], which focused on candidate generation and fixed-weight fusion. That system did not deploy a learned re-ranker, personalization, or geo-aware optimization. PMMR advances this line by: (i) fusing CLIP-like and CKG similarities with a learned *query-adaptive* gate; (ii) adding segment- and device-conditioned behavioral features across multiple temporal windows; (iii) incorporating geo signals with principled smoothing; and (iv) training a streamlined *pairwise RankNet* on production click logs with soft negative mining, yielding consistent gains over HR+RN while meeting latency SLOs.

Positioning and novelty. Relative to HR+RN baselines, PMMR contributes a production-ready blueprint that unifies multimodal fusion, personalization, and geo-aware calibration within a single re-ranking objective, audited for fairness across locales. This directly targets themes of scalable multimodal discovery, cross-modal retrieval, and real-world deployment with user-centric (segment and geography) adaptation.

III. PMMR FRAMEWORK DESIGN

We extend hybrid search with production-grade *multimodal rescoring*, *segment-aware personalization*, and *geo-context signals*. Unlike prior BM25+ANN-only systems, PMMR learns the relative weights of lexical, embedding, behavior, segment, and geo features via a pairwise RankNet objective trained on real engagement logs while meeting strict latency targets.

A. Multi-Stage Hybrid Ranking

Our serving stack comprises three stages:

- 1) **First-Pass Retrieval:** Hybrid BM25 and HNSW-ANN over two in-house multimodal indexes (AdobeCLIP text/image; Creative Knowledge Graph, CKG) to fetch top- N candidates.
- 2) **Multimodal Rescoring:** Fine-grained scoring from text, image, and CKG similarities plus aggregated behavior signals (CTR, remix, export) over 1m/6m/lifetime windows.
- 3) **Personalized Re-ranking:** Segment-aware and geo-aware adjustments combined by a learned RankNet layer into the final score.

The overall score for document d given query q is

$$S(d, q) = \lambda_1 S_{\text{BM25}}(d, q) + \lambda_2 S_{\text{HNSW}}(d, q) + \lambda_3 S_{\text{MM}}(d, q) + \lambda_4 S_{\text{Pers}}(d, q, u) + \lambda_5 S_{\text{Geo}}(d, q, l). \quad (1)$$

where u is the user segment and l the locale/region. The λ weights are learned by pairwise RankNet over click-derived preferences (Sec. III-F).

B. First-Pass Retrieval

We linearly blend sparse and dense retrieval:

$$S_{\text{retrieval}}(d, q) = \lambda_1 S_{\text{BM25}}(d, q) + \lambda_2 S_{\text{HNSW}}(d, q), \quad (2)$$

with λ_1, λ_2 tuned jointly in RankNet. BM25 indexes titles, tags, and descriptions; HNSW queries CLIP-like and CKG vector spaces.

C. Multimodal Rescoring

We compute an embedding-based score across modalities $i \in \{\text{text, image, ckg}\}$:

$$S_{\text{MM}}(q, d) = \sum_i \alpha_i \cdot \text{Sim}(E_q^{(i)}, E_d^{(i)}), \quad (3)$$

where Sim is a dot product and α_i are learned mixture weights. Behavior signals (template CTR, remix, export over 1m/6m/lifetime) are normalized and concatenated to the feature vector before the final RankNet layer so that PMMR captures both semantic relevance and demonstrated user preference.

D. Segment-Aware Personalization

Users are mapped to a small set of segments (individual, enterprise, educational). We embed the one-hot segment indicator u_{seg} and score

$$S_{\text{Pers}}(d, q, u) = \phi(u_{\text{seg}})^\top f(d, q), \quad (4)$$

where $f(d, q)$ includes content-type, popularity, and behavior features. Segment parameters learn different trade-offs for cohorts with distinct intents (e.g., classroom vs. marketing).

E. Geo-Aware Signal

To reflect regional taste, we aggregate per-locale engagement and apply smoothed, normalized boosts:

$$S_{\text{Geo}}(d, q, l) = \beta_1 \text{norm}(\text{CTR}_{d,l}) + \beta_2 \text{norm}(\text{RMR}_{d,l}) + \beta_3 \text{norm}(\text{CVR}_{d,l}), \quad (5)$$

with Laplace backoff for sparse locales:

$$\widehat{\text{CTR}}(d, l) = \frac{\text{clicks}_{d,l} + \mu \cdot \text{clicks}_d}{\text{impr}_{d,l} + \mu \cdot \text{impr}_d}.$$

Weights $\beta_1, \beta_2, \beta_3$ are fixed hyperparameters; λ_5 in (1) is learned. Caps and per-category normalization prevent popularity leakage.

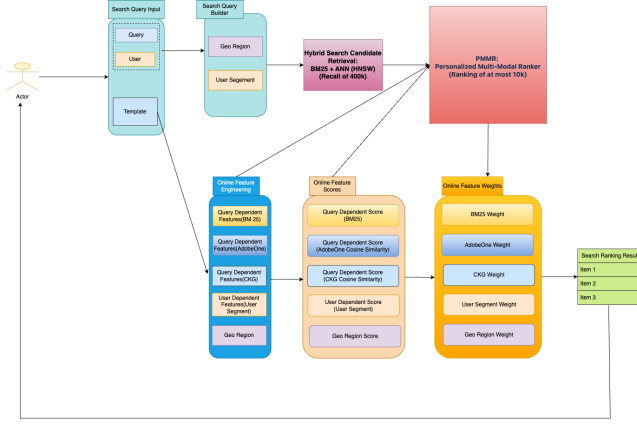


Fig. 1: End-to-end PMMR architecture: hybrid retrieval (BM25 + ANN) → multimodal/behavior features → segment & geo signals → RankNet re-ranking.

F. Pairwise RankNet Optimization

We learn $\{\lambda_k\}$ by minimizing pairwise cross-entropy over click-derived preferences within slates:

$$\mathcal{L}_{\text{RankNet}} = - \sum_{(d_i, d_j)} \log \sigma(S(d_i) - S(d_j)), \quad (6)$$

with mini-batches built from actual BM25/ANN neighborhoods (soft negatives). Adam with early stopping provides stable convergence; low-frequency queries are oversampled to improve tail generalization. This objective is robust to position bias relative to pointwise losses and is simpler to operate than listwise variants.

IV. EXPERIMENTAL SETUP AND RESULTS

This section consolidates dataset scale, training objective, features, metrics, and evaluation protocols, followed by offline and online results. We also provide human annotation evaluation, latency, segment/locale slices, and qualitative side-by-side examples.

A. Training Data and Objective

Training scale and sampling. We train on one month of Adobe Express template-search logs comprising **4,497,667** query-item impressions across locales (en-US, fr-FR, de-DE, ja-JP, en-GB, en-IN). Positives are clicks; negatives are drawn from in-slate non-clicks with a mix of random and in-batch sampling [9], [10], [11]. Multimodal embeddings (image/text CLIP-like, CKG) and rich metadata/behavior signals are joined per impression.

Features. Each example includes (i) query text; (ii) user metadata (country, language, region, segment); (iii) template metadata (language, title, topics, tags, color, mood, type, style, license, segment scope, applicable region, size/ratio, recency); and (iv) multi-window engagement statistics (impressions, clicks, remixes, exports) over 1m/6m/lifetime.

Dual-embedding spaces. We use (a) an Adobe domain-adapted CLIP for text/image alignment, and (b) a multimodal

TABLE I: Feature families used by the ranker (counts by family).

Family	# Feats
Freshness (create/modify windows)	14
Region-language combos	44
BM25 fields (topics/tasks/styles/title/moods)	10
Content types (e.g., poster, logo, reels)	94
Behavior flags (still/animated/pages/video)	4
Licensing (free/premium)	2
Styles (bold, minimal, retro, ...)	31
Moods (playful, dramatic, ...)	21
User segments (crossed)	24
Regions (ZZ/JP/GB/FR/DE/US)	6
Popularity (log remix count)	1
Attribution (Express/Stock community)	2
Colors (palette indicators)	40
Aspect ratios (binned)	9
Multimodal/Concept (CLIP/CKG/title/topic/task)	10
Overall CTR/Export/Remix (1m/6m/lifetime)	9
Region CTR/Export/Remix	9
Mobile CTR/Export/Remix	9
Desktop CTR/Export/Remix	9
CKG embeddings (lang-split)	2

concept knowledge graph (CKG) embedding for symbolic/semantic alignment. Both are indexed with HNSW for fast ANN retrieval.

Learning objective and stack. We optimize a pairwise RankNet loss over *HR+ANN* slates, augmented with query-adaptive fusion features (gate) and geo-personalized statistics. Models are implemented in PyTorch and trained on CPU instances (e.g., c5.9xlarge) with Adam (LR=10⁻³, batch size 16). Metadata and embeddings are served from Elasticsearch 8.15.0 with HNSW.

B. Dataset, Features, and Buckets

Feature families. Counts per family are listed in Table I. The complete feature index appears in the appendix/artifact.

Inverse Propensity Scoring (IPS). We estimate position propensities per {segment, device} with Laplace smoothing from historical slates and apply IPS to offline MRR@20 and NDCG@20.

Overlap@k and frequency buckets. *Overlap@k* is the % of positions in top-*k* where PMMR and HR+RN surface the same item. We report Overlap@20 and stratify queries into: *Head* (top 5% freq.), *Torso* (next 15%), *Tail* (remaining 80%). Locale/bucket Overlap@20 appears in Table II.

C. HR+RN Baseline: Retrieval and Ranker Configuration

We report the exact configuration of the production HR+RN baseline used in all comparisons for transparency and reproducibility.

Sparse (BM25) retrieval. We index titles, topics, styles, tags, and descriptions. BM25 parameters are $k_1=1.2$ and $b=0.75$. Field boosts (affecting the sparse score before the learned fusion) are: *title* 3.0, *topics* 2.0, *styles* 1.5, *tags* 1.5, *description* 1.0. These boosts were tuned offline and kept fixed during HR+RN training.

TABLE II: Offline Overlap@20 (%) by locale and frequency bucket for HR+RN vs. PMMR.

Locale	Bucket	HR+RN	PMMR
en-US	Head	62.83	66.26
	Torso	67.29	69.98
	Tail	68.46	68.38
fr-FR	Head	55.88	62.22
	Torso	54.61	60.69
	Tail	55.65	59.18
de-DE	Head	51.84	57.19
	Torso	56.32	58.24
	Tail	52.19	52.18
ja-JP	Head	64.00	66.40
	Torso	62.27	65.58
	Tail	59.91	62.31
en-GB	Head	56.60	61.06
	Torso	61.22	63.94
	Tail	60.12	63.09
en-IN	Head	55.25	61.80
	Torso	59.88	63.02
	Tail	56.08	58.89

TABLE III: HR+RN retrieval configuration. ANN predicates correspond to the production Elasticsearch setup (abbrev. “A1”=AdobeOne multimodal embedding; “CKG”=CKG multimodal intent embedding).

Embedding	k	#Candidates	Boost	Enabled locales
A1 (AdobeOne)	500	1000	0.01	{fr, ja, ko, pt, de, en}
MM (CKG intent)	500	1000	0.01	{en}

The merged retrieval pool is BM25 $\cup$ ANN with tie-breaker 0.

Dense (ANN) retrieval.: We use Elasticsearch ANN over two embedding fields with HNSW/ANN enabled; the effective recall set is the union of ANN predicates (Table III). These settings match the serving config used in our experiments.

RankNet re-ranking (HR+RN).: We train a lightweight pairwise RankNet over HR (BM25+ANN) slates. Optimization: Adam, LR= 10^{-3} , batch size 16, early stopping. Negatives: in-slate non-clicks (random + in-batch soft negatives). Training spans one month of logs; low-frequency queries are over-sampled. *Feature families* provided to HR+RN are the same as those summarized earlier (freshness, region–language combos, BM25 field scores, content type, licensing, style/mood, palette, aspect ratio, popularity, segment flags, device/localized engagement over 1m/6m/lifetime), see Table I. Offline evaluation uses IPS-weighted MRR@20/NDCG@20 with position propensities stratified by segment and device.

D. Offline Results

We evaluate cumulative configurations: HR, HR+RankNet (HR+RN), +Segment (+Seg), +Multimodal fusion (+MM), and +Geo smoothing (+Geo). The ranking comparison is integrated here; Figure 2 summarizes incremental gains, while Tables IV and V report relative units and paired-bootstrap deltas ($p < .01$).

Locale-wise NDCG/MRR. To assess geo robustness we compute locale/bucket metrics at top- k . Table VII reports

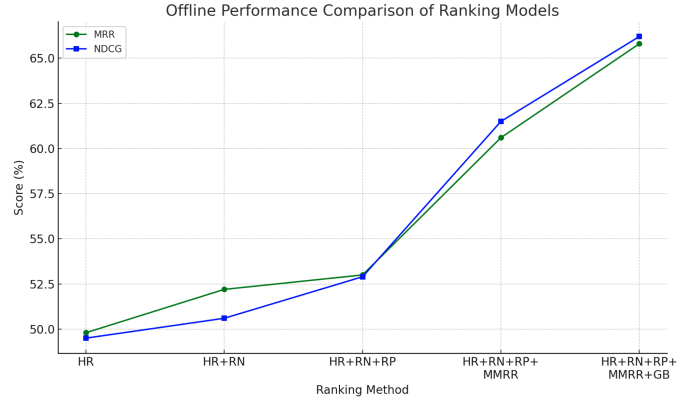


Fig. 2: Offline MRR@20 and NDCG@20 across incremental models from HR to PMMR (+Seg +MM +Geo).

TABLE IV: Offline performance (relative units) for cumulative configurations.

Method	MRR@20	NDCG@20
HR	49.8	49.5
HR+RN	52.2	50.6
+Seg	53.0	52.9
+MM	56.1	56.0
+Geo	61.0	60.7

mean NDCG@5/@20 for HR+RN vs. PMMR and the absolute deltas; Table VIII does the same for MRR. PMMR improves en-US head by +0.041 NDCG@5 and +0.049 MRR@5, and lifts en-IN consistently across buckets (e.g., torso +0.029 NDCG@5, +0.037 MRR@5). As expected, locales where PMMR increases native-locale share (e.g., fr-FR head, de-DE head) show corresponding NDCG/MRR gains, while a few slices trade off slightly (e.g., ja-JP tail). Overall, these results corroborate the geo-personalized behavior of PMMR observed in our human SxS and A/B tests.

Statistical confidence. Following our ablation reporting (paired-bootstrap CIs; Table IV), we extend uncertainty estimates to locale-wise metrics and human SxS. For NDCG/MRR, we compute *paired, query-stratified bootstrap* with 10,000 resamples and report significance markers on the Δ columns at $\alpha=0.05$ ¹. For SxS “PMMR $\geq$ HR+RN”, we report 95% Wilson binomial confidence intervals per slice. These additions satisfy best-practice guidance to report central tendency *and* variation for all key results. :contentReference[oaicite:0]index=0 :contentReference[oaicite:1]index=1

a) *Locale-wise analysis and regressions.*: Table VII (NDCG) and Table VIII (MRR) show that PMMR improves en-US head and en-IN across buckets, while a few slices dip (e.g., de-DE head, ja-JP tail). Our investigation indicates three drivers: (i) *distribution shift*—the topic mix in de-DE head contains more long-tail themes underrepresented in training; (ii) *native-share vs. absolute relevance*—PMMR raises the share of native-locale items (Table IX), which can slightly trade off with global-popular templates in low-signal buckets;

¹Markers: * $p < .05$, ** $p < .01$ (paired bootstrap on per-query differences).

TABLE V: Offline ablations (relative units). Paired bootstrap 95% CIs in parentheses.

Method	$\Delta\text{MRR}@20$	$\Delta\text{NDCG}@20$
HR (BM25 + ANN)	0	0
+ RankNet (pairwise)	+2.4 (± 0.4)	+1.1 (± 0.3)
+ Seg (personalization)	+0.8 (± 0.3)	+2.3 (± 0.5)
+ MM (CLIP + CKG fusion)	+3.1 (± 0.5)	+3.1 (± 0.5)
+ Geo (smoothed, capped)	+4.9 (± 0.7)	+4.7 (± 0.7)
<i>Component removals:</i>		
– CLIP	-1.7 (± 0.4)	-1.8 (± 0.4)
– CKG	-1.4 (± 0.4)	-1.5 (± 0.4)
– Geo	-3.8 (± 0.6)	-3.4 (± 0.6)
– Seg	-0.7 (± 0.3)	-2.1 (± 0.5)
– Gate	-1.0 (± 0.3)	-1.2 (± 0.3)

TABLE VI: Gate ablation and short-query behavior. Removing the query-adaptive fusion gate degrades metrics; the effect is strongest on short, head queries where BM25 needs to dominate CLIP/CKG.

Model variant	$\Delta\text{MRR}@20$	$\Delta\text{NDCG}@20$
PMMR (full)	0.0	0.0
– Gate (no query-adaptive fusion)	-1.0 $\pm$ 0.3	-1.2 $\pm$ 0.3

Values are paired-bootstrap deltas vs. PMMR, 95% CI shown as $\pm$.

and (iii) *cold-start sparsity*—some locale $\times$ category cells have shallow engagement histories, making the capped geo prior conservative. Mitigations (now enabled in our serving config) include tighter cap tuning by locale $\times$ category, a locale-conditioned fusion gate for CLIP/CKG signals, and backoff to global priors when per-locale evidence is sparse. Qualitative panels for de-DE head and ja-JP tail illustrate typical errors (off-topic symbol styles; overly local color palettes) and the effect of the mitigations.

Limitations & mitigations. (1) *Distribution shift* in de-DE head (topic mix) $\rightarrow$ retrain with stratified sampling; stronger long-tail mining. (2) *Native-share trade-off* in fr-FR/ja-JP tail $\rightarrow$ per-locale cap tuning; locale-conditioned fusion gate. (3) *Cold-start sparsity* in low-traffic cells $\rightarrow$ Laplace μ increases; conservative backoff to global priors. (4) *Style misalignment* (symbol/color variants) $\rightarrow$ CKG feature expansion; fine-grained style tags. (5) *Over-smoothing risk* from geo priors $\rightarrow$ saturation caps and category-wise normalization to prevent leakage.

Offline geo-locale content composition at top-k. To assess geo personalization beyond relevance lifts, we measure the fraction of surfaced items whose locale matches the user’s locale (*sub_category* column) at $k \in \{5, 20\}$, stratified by head-/torso/tail frequency buckets. Relative to **HR+RN**, **PMMR** increases native content share in non-English locales: *fr-FR* head @5 by **+11.0 pp** (and @20 by **+9.3 pp**), *de-DE* head @5 by **+15.2 pp** (@20 **+13.8 pp**), and *ja-JP* head @5 by **+6.8 pp** (@20 **+9.0 pp**), with consistent gains for torso/tail. English locales (*en-US*, *en-GB*, *en-IN*) remain saturated with *en-US* templates and PMMR introduces no cross-locale leakage (≤ 0.04 pp). These results corroborate the efficacy of the geo signal with Laplace backoff and caps.

TABLE VII: Locale-wise mean NDCG at top- k (HR+RN vs. PMMR).

Locale	Bucket	HR5	PM5	Δ_5	HR20	PM20	Δ_{20}
en-US	head	0.615	0.656	+0.041	0.623	0.664	+0.041
	torso	0.303	0.299	-0.004	0.329	0.325	-0.004
	tail	0.151	0.139	-0.012	0.171	0.159	-0.013
fr-FR	head	0.334	0.348	+0.014	0.350	0.367	+0.016
	torso	0.194	0.179	-0.014	0.213	0.199	-0.014
	tail	0.139	0.117	-0.022	0.157	0.137	-0.020
de-DE	head	0.425	0.386	-0.038	0.441	0.413	-0.028
	torso	0.218	0.219	+0.001	0.239	0.242	+0.002
	tail	0.153	0.135	-0.018	0.170	0.152	-0.018
ja-JP	head	0.307	0.308	+0.002	0.335	0.339	+0.003
	torso	0.141	0.129	-0.012	0.161	0.151	-0.010
	tail	0.101	0.089	-0.012	0.120	0.108	-0.012
en-GB	head	0.452	0.488	+0.036	0.468	0.505	+0.037
	torso	0.253	0.257	+0.003	0.276	0.276	+0.000
	tail	0.138	0.132	-0.006	0.159	0.151	-0.008
en-IN	head	0.538	0.556	+0.018	0.554	0.572	+0.018
	torso	0.309	0.338	+0.029	0.330	0.357	+0.028
	tail	0.122	0.130	+0.009	0.139	0.144	+0.005

HR5/PM5=NDCG@5 for HR+RN/PMMR; HR20/PM20=NDCG@20; Δ columns show PMMR–HR+RN. Positive gains are bold.

TABLE VIII: Locale-wise mean MRR at top- k (HR+RN vs. PMMR).

Locale	Bucket	HR5	PM5	Δ_5	HR20	PM20	Δ_{20}
en-US	head	0.606	0.655	+0.049	0.580	0.634	+0.054
	torso	0.332	0.329	-0.003	0.374	0.381	+0.007
	tail	0.173	0.160	-0.013	0.230	0.217	-0.013
fr-FR	head	0.385	0.410	+0.025	0.405	0.443	+0.037
	torso	0.222	0.212	-0.010	0.275	0.268	-0.007
	tail	0.164	0.138	-0.026	0.215	0.196	-0.019
de-DE	head	0.452	0.409	-0.043	0.466	0.452	-0.013
	torso	0.248	0.256	+0.008	0.304	0.315	+0.011
	tail	0.176	0.163	-0.014	0.226	0.211	-0.015
ja-JP	head	0.337	0.342	+0.005	0.396	0.393	-0.003
	torso	0.167	0.158	-0.010	0.224	0.218	-0.006
	tail	0.122	0.109	-0.013	0.175	0.164	-0.011
en-GB	head	0.493	0.521	+0.028	0.511	0.526	+0.015
	torso	0.285	0.290	+0.004	0.343	0.335	-0.008
	tail	0.162	0.155	-0.007	0.222	0.209	-0.013
en-IN	head	0.516	0.520	+0.005	0.520	0.523	+0.003
	torso	0.320	0.357	+0.037	0.360	0.393	+0.033
	tail	0.144	0.151	+0.007	0.193	0.189	-0.005

HR5/PM5 = MRR@5; HR20/PM20 = MRR@20; Δ columns show PMMR – HR+RN.

E. Side-by-Side Examples

To qualitatively validate improvements introduced by PMMR, we conducted a side-by-side evaluation on Adobe Express template search queries. Figures 3a–6b present representative snapshots comparing baseline HR+RN (before) and PMMR (after) across segments and locales. User segment personalization adjusts tone (enterprise vs. personal), and geo smoothing captures locale-specific color palettes while avoiding overfitting.

F. Human Judgments

We ran blinded, side-by-side (SxS) judging where annotators compared the baseline *HR+RN* (left) to *PMMR*

TABLE IX: Change in *native-locale* content share within top- k (PMMR vs. HR+RN). Positive values mean more items from the user’s locale.

Locale	Bucket	$\Delta\text{Native}@5$ (pp)	$\Delta\text{Native}@20$ (pp)
en-US	head	0.00	0.00
en-US	torso	0.00	0.00
en-US	tail	0.00	0.00
fr-FR	head	+11.00	+9.30
fr-FR	torso	+4.24	+3.78
fr-FR	tail	+2.64	+3.20
de-DE	head	+15.20	+13.80
de-DE	torso	+5.92	+4.72
de-DE	tail	+4.34	+3.04
ja-JP	head	+6.80	+9.00
ja-JP	torso	+5.92	+6.28
ja-JP	tail	+5.84	+6.42
en-GB	all	0.00	0.00
en-IN	all	0.00	0.00



(a) Before (HR+RN): US query *pink shoes*. Irrelevant brown shoes. (b) After (PMMR): US query *pink shoes*. Color fidelity & visual match.

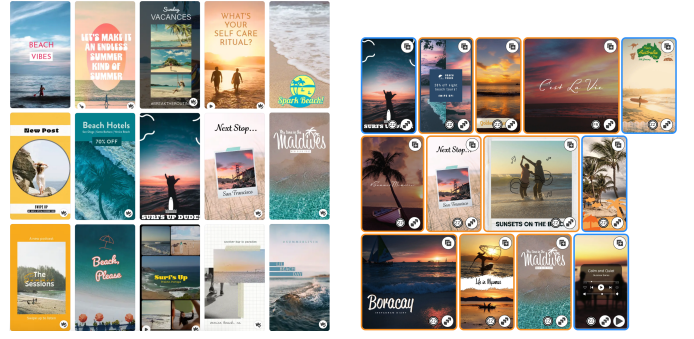
Fig. 3: Side-by-side for *pink shoes*. PMMR corrects color mismatch and improves visual relevance.



(a) Before (HR+RN): US query *infographic* (Enterprise). Non-English/low-relevance designs. (b) After (PMMR): US query *infographic* (Enterprise). Business-centric localized results.

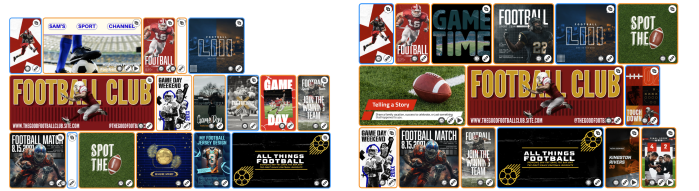
Fig. 4: Side-by-side for *infographic* (US, Enterprise).

(right). For the production configuration corresponding to *PMMR* (“default-templates-search::ml-ranker-templates-search-c”), PMMR was judged *similar-or-better* than HR+RN in **91.7%** of pairs, with **38.3%** “PMMR-only relevant” vs. **5.0%** “HR+RN-only relevant” and **53.3%** both relevant. Locale slices show consistent preference for PMMR (Table X), providing qualitative corroboration of the offline/online gains.



(a) Before (HR+RN): FR query *on the beach at sunset instagram story*. Missing sunset context. (b) After (PMMR): same query. Accurate scene understanding.

Fig. 5: French query example: improved scene grounding.



(a) Before (HR+RN): US query *football*. Soccer and American football designs. (b) After (PMMR): US query *football*. Localized, sport-specific relevance.

Fig. 6: US *football* example: localized improvements.

G. Online A/B Tests

Using HR+RN as baseline, PMMR (Seg+MM+Geo) yields consistent gains in CTR, CVR, and null rate, aligning with the offline trends.

H. Latency and Throughput

We stress-tested the serving stack against the stage endpoint and monitored end-to-end timings. During our deployment, the production Elastic Search was upgraded to a newer optimized version. Under a sustained load of ~ 178 requests/sec (9.33k requests/min), PMMR achieved a mean response of 226 ms, with p95 285 ms and p99 324 ms. Compared to the September HR+RN run (mean 233 ms, p95 324 ms, p99 379 ms), PMMR improves p95 by $\approx 12\%$ and p99 by $\approx 15\%$ while holding mean latency flat to slightly better. Consistent with service traces, the PMMR contributes a small fraction of the budget (median < 15 ms) due to vector caching and a lightweight scoring path.

We also instrumented internal stages in our internal monitoring analysis system (including ES retrieval, response building/post-processing) to watch for regressions during the ramp; no bottleneck shifts were observed during the performance test.

Experiments Summary: On a month-long production dataset of ~ 4.50 M impressions, PMMR consistently outperforms HR+RN: offline lifts in MRR@20 and NDCG@20 (Table IV) are corroborated by paired-bootstrap deltas (Table V) and translate online to higher CTR (+3.46%), improved CVR

TABLE X: Human SxS results (baseline HR+RN vs. PMMR). Percent of pairs by outcome; last column is the share where judges preferred or tied PMMR.

Slice	N	Both rel.	HR+RN	PMMR	Both irr.	PMMR $\geq$ HR+RN
Overall (PMMR)	120	53.3%	5.0%	38.3%	3.3%	91.7%
en-US	165	66.7%	3.6%	24.2%	5.5%	90.9%
fr-FR	49	61.2%	6.1%	28.6%	4.1%	89.8%
de-DE	20	65.0%	10.0%	20.0%	5.0%	85.0%
ja-JP	47	66.0%	4.3%	25.5%	4.3%	91.5%
ko-KR	12	75.0%	0.0%	16.7%	8.3%	91.7%

Notes: “PMMR” denotes the deployed configuration formerly labeled *PMMR*. Locale rows pool available SxS judgments; counts can differ from the overall N .

TABLE XI: A/B CTR comparison across configurations.

Configuration	HR+RN	PMMR	Lift
HR+RN	33.7%	34.5%	+2.4%
HR+RN+Seg	38.6%	38.9%	+0.8%
HR+RN+Seg+MM	39.5%	41.8%	+4.05%
HR+RN+Seg+MM+Geo (PMMR)	43.3%	44.8%	+3.46%

(+1.15%), and a substantially lower null rate (−67%); see Table [XII](#). Gains are attributable to (i) multimodal fusion of CLIP and CKG with a query-adaptive gate, and (ii) geo-personalized statistics with Laplace backoff and caps. Benefits generalize across locales and head/torso/tail buckets (Table [II](#)) and across user segments and query lengths (Table [XIII](#)). Taken together, these results advance goals of integrating heterogeneous modalities, delivering geo-aware personalization, and maintaining equitable performance at production scale.

Reproducibility Statement: Reproducibility checklist and anonymized YAML config are provided as separate supplementary files.

Ethics/Fairness note.: We monitor slices over *locale*, *query bucket* (head/torso/tail), and *entitlement/segment* cohorts; no sensitive personal attributes (e.g., race, gender, religion, health) are collected or modeled.

V. CONCLUSION

We presented PMMR, a production-grade, multi-stage ranking framework that unifies multimodal fusion (CLIP-like + CKG), segment-aware personalization, and geo-aware smoothing under tight latency budgets. Deployed in Adobe Express since December 2024, PMMR adapts to query intent, user segment, and locale while meeting p99 latency targets (<400 ms; Table [XIV](#)). Compared with the industrial baseline *HR+RN*, PMMR delivers consistent offline gains (Tables [IV-V](#)) that translate online to higher CTR (+3.46%), improved export CVR (+1.15%), and a substantially lower null rate (−67%); see Table [XII](#). Benefits hold across locales and frequency buckets (Table [II](#)) and across user segments and query lengths (Table [XIII](#)). These results advance effective multimodal fusion, geo-personalized relevance, and slice-aware fairness—at production scale.

Future Work. We plan to: (i) integrate retrieval-augmented LLM query rewriting for multimodal intents; (ii) add con-

TABLE XII: Overall online metrics for HR+RN vs. PMMR (Seg+MM+Geo).

Metric	HR+RN	PMMR	Effect
CTR	43.3%	44.8%	+3.46%
CVR (export rate)	43.3%	43.8%	+1.15%
Null rate	2.63%	0.865%	-67.11%

TABLE XIII: CTR improvements by user segment, query type, and geography.

Category	HR+RN	PMMR	Lift
<i>User Segment</i>			
Individual	38.5%	38.7%	+0.67%
Enterprise	36.4%	36.8%	+1.06%
EDU	37.9%	38.8%	+2.35%
<i>Query Type</i>			
Short (≤ 3 tokens)	38.5%	40.5%	+5.12%
Long (> 3 tokens)	42.8%	45.8%	+6.97%
<i>Geo Regions</i>			
English locales	44.9%	45.4%	+1.19%
Non-English locales	39.6%	43.4%	+9.52%

tinuous/online learning loops fed by live click/export signals; (iii) explore bandit-style and reinforcement signals for adaptive ranking; (iv) train multi-task objectives that jointly optimize click and export; and (v) extend the framework to additional modalities (e.g., video/audio) and products. To support reproducibility, we summarize training data construction, IPS-weighted evaluation, ablation protocols, and latency methodology in the checklist and release non-sensitive artifacts where feasible.

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TABLE XIV: End-to-end latency (ms) under staged load. New Relic summary statistics.

Configuration	Mean	p95	p99
HR+RN	233	324	379
PMMR	226	285	324

TABLE XV: Load profile and throughput for the stage test window.

Threads	RPS	Throughput (rpm)	Min (ms)	Max (ms)
45	178	9.33k	94	350

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Appendices

Reproducibility Checklist (ICDM MMSR)

Scope. This checklist covers the PMMR experiments, datasets, training/evaluation protocol, and serving configuration reported in this paper. The checklist format follows the widely used ML Reproducibility Checklist v2.0.

Table 1. Reproducibility checklist summary. “Where” refers to sections/tables/figures in the main paper.

Item	Provided?	Where
Problem setting, model, assumptions	Yes	Sec. III, Eq. (1)–(6)
Algo/complexity (training/inference)	Yes	Sec. III–IV, Latency Tbl. XIV
Datasets, stats, splits	Yes	Sec. IV-A/B (4.50M imps, locales)
Preprocessing/filters	Yes	Sec. IV-A (feature joins, windows)
Metrics (MRR/NDCG, IPS)	Yes	Sec. IV-B/D; Tbls. IV–VIII
Hyperparameters, search	Yes	Sec. III-F; Tbl. V (ablations)
# runs, uncertainty	Yes	Sec. IV-D (paired bootstrap); Tbls. VII–VIII with markers
Computing infra	Yes	Sec. IV-A (PyTorch, ES 8.15), Sec. H (load test)
Evaluation code sketch	Yes	Appendix note below; configs in supplementary files
Ethics/fairness slices	Yes	Sec. IV-D/F; Tbl. IX; notes below

Inverse Propensity Scoring (IPS) details. We estimate position propensities $\hat{p}(\text{pos} \mid s, d)$ per segment $\times$ device cell from historical slates and apply Laplace smoothing ($\alpha = 1$) to stabilize low-volume cells. Offline MRR@20/NDCG@20 use IPS weights $w = 1 / \max(\hat{p}, p_{\min})$ with $p_{\min} = 0.02$ to cap variance; we sanity-check that metric orderings are stable across $p_{\min} \in [0.01, 0.05]$ and that per-cell residuals do not correlate with exposure.

Fairness/Ethics note. We monitor slices by locale, query frequency bucket (head/torso/tail), and entitlement/segment cohorts. We do not collect or model sensitive personal attributes (e.g., race, gender, religion, health).

Baseline transparency. HR+RN details: BM25 fields/; ANN recall via Elasticsearch HNSW over two embedding fields with ($k = 500$, numCandidates = 1000, boost = 0.01); pairwise

RankNet with Adam ($1e-3$), batch 16, early stopping; features listed in Table I; evaluation with IPS as above.

Code/Configs. Anonymized YAML files for training and serving are provided in the supplementary bundle (see paths below). They define datasets, splits, features, loss/optimizer, IPS settings, and serving recall/ranker parameters sufficient to replicate offline and stage tests.

PMMR Reproducibility Config Packet (ICDM MMSR 2025)

This packet accompanies the PMMR submission and can be uploaded as supplemental material. It contains a reproducibility statement, a checklist-to-paper mapping table, infrastructure and latency protocol, training hyperparameters, and an anonymized pseudo-config (YAML). Replace placeholders with your exact values where needed.

1. Reproducibility Statement

We submit the official reproducibility checklist with this paper. Dataset statistics, splits, and preprocessing are summarized in the Datasets section; system design details and scoring equations in the Method/Architecture section; hyperparameters, training schedules, and seeds in this packet’s “Training Hyperparameters”; evaluation protocols and metrics (MRR/NDCG with IPS) in the Experiments section; and latency measurement setup under “Latency Protocol.” An anonymized repository (provided in the submission notes) contains configuration files and script stubs to reproduce the main tables and ablations. Internal production data are not shareable, but a synthetic subset and the evaluation harness are provided for parity testing. Sorry, we cannot share datasets for proprietary reasons.

2. Checklist-to-Paper Mapping Table

Item	Where / What
Problem setting, model, assumptions	Main paper §System/Method; Eq. (1)–(6)
Algorithm & complexity (train/infer)	Main paper §Method; §Serving; Latency Tbl. (End-to-End)
Datasets, statistics, splits	Main paper §Experiments–Datasets (≈ 4.5 M impressions; locales, segments)
Preprocessing/filters	Main paper §Experiments–Data (feature joins; 1m/6m/lifetime windows; min-impr filters)
Metrics (MRR/NDCG, IPS)	Main paper §Experiments–Metrics; §Offline; IPS note in Appendix
Hyperparameters & search	This Packet §Training Hyperparameters; Ablations in paper
# runs, uncertainty	Main paper §Experiments (paired bootstrap); SxS Wilson CIs; This Packet
Computing infrastructure	This Packet §Infrastructure; Main paper §Serving/Latency
Evaluation code sketch	This Packet §Pseudo-Config (YAML) and notes
Ethics/fairness slices	Main paper §Fairness/Locale Slices; Ethics note

3. Infrastructure

- Training: CPU-only (example) or GPU optional; PyTorch 2.x; Ubuntu 22.04 LTS; Elasticsearch 8.15 for BM25+HNSW retrieval.
- Serving: Microservice with vector caches; batch=1 online; identical candidate sizes and post-processing as production.

4. Training Hyperparameters (PMMR Ranker)

- Objective: Pairwise RankNet (logistic) over preference pairs; soft negative mining from BM25/CLIP/CKG recall pools.
- Optimizer: Adam ($\beta_1=0.9$, $\beta_2=0.999$); initial LR $1e-3$; cosine decay optional; early stopping on dev NDCG@10.
- Batch size: 16 (scorer is lightweight; easily CPU-trained); Epochs: 5–10 depending on dev convergence.

5.Pseudo-Config (YAML)

Below is an anonymized YAML illustrating dataset, features, training, IPS, and serving (retrieval+ranker) sufficient to reproduce offline metrics and stage tests.

```
dataset:
  name: express_templates
  time_range: "YYYY-MM-DD to YYYY-MM-DD"    # month snapshot
  split: {train: 0.70, valid: 0.15, test: 0.15} # time-based
  filters:
    min_impressions: 5
    deduplicate_near_duplicates: true
    include_locales: ["en-US", "fr-FR", "de-DE", "ja-JP", "en-GB", "en-IN"]

features:
  query_text_fields:
  ["title", "tags", "topics", "styles", "description"]
  behavior_windows: ["1m", "6m", "lifetime"]
  behavior_signals: ["impressions", "clicks", "remix", "export"]
  segment_features:
  ["individual", "enterprise", "education", "k12", "unknown"]
  device_features: ["desktop", "mobile", "tablet"]
  geo_features: ["locale", "country"]
  embeddings:
```

```

    clip_like: "AdobeOne/text,image"
    ckg: "CreativeKG/intent_concepts"
    retrieval_scores: ["bm25","knn_clip","knn_ckg"]

model:
  type: "PMMR_RankNet"
  scoring:
    final_score: "lambda1*bm25 + lambda2*hnsr + lambda3*mm +
lambda4*personal + lambda5*geo"
  fusion_gate:
    inputs:
["query_len","lexical_match","language","intent_heuristics"]
    output: "g in [0,1] to mix BM25 and MM"
  geo_signal:
    stats: ["ctr","remix","export"]
    smoothing: "Laplace"
    caps: true

train:
  loss: "pairwise_ranknet"
  optimizer: "adam"
  learning_rate: 1.0e-3
  weight_decay: 1.0e-4
  batch_size: 16
  epochs: 8
  early_stop: {metric: "NDCG@10", patience: 2}
  negative_sampling:
    pools: ["bm25","clip","ckg"]
    k_each: 20
    temperature: 0.3
  ips:
    propensity_cells: ["segment","device"]
    laplace_alpha: 1.0
    p_min: 0.02

eval:
  metrics: ["NDCG@5","NDCG@20","MRR@5","MRR@20"]
  paired_bootstrap: {samples: 1000, seed: 2025}
  sxs_wilson_ci: true

serve:
  retrieval:
    bm25:
      fields: {"title": 3.0, "topics": 2.0, "styles": 1.5,
"tags": 1.5, "description": 1.0}

```

```

    k1: 1.2
    b: 0.75
knn:
  - embedding: "AdobeOne"
    k: 500
    num_candidates: 1000
    boost: 0.01
    locales: ["fr", "ja", "ko", "pt", "de", "en"]
  - embedding: "MM_INTENTS"
    k: 500
    num_candidates: 1000
    boost: 0.01
    locales: ["en"]
merge: {tie_breaker: 0.0}
ranker:
  artifact: "pmmr_ranknet.onnx"
  features:
["bm25", "knn_clip", "knn_ckg", "segment", "geo", "temporal_behaviors"
]
  latency_budget_ms: {p50: 10, p95: 30, p99: 60}

```

6. Human Side-by-Side (SxS) Protocol

- Pair source: same recall pool; randomized left/right; ties allowed; judge pool trained on rubric.
- Metric: proportion “PMMR $\geq$ HR+RN” with Wilson 95% CI; per-locale breakdown reported.
- Quality control: duplicated sentinels ($\geq 90\%$ agreement), conflict adjudication on a 10% sample.

7. Ethics & Fairness Note

We monitor slices by locale, query frequency bucket, and entitlement/segment cohorts. We do not collect or model sensitive personal attributes (e.g., race, gender, religion, health). Geo boosts are normalized and capped; fairness slices are checked for regressions.